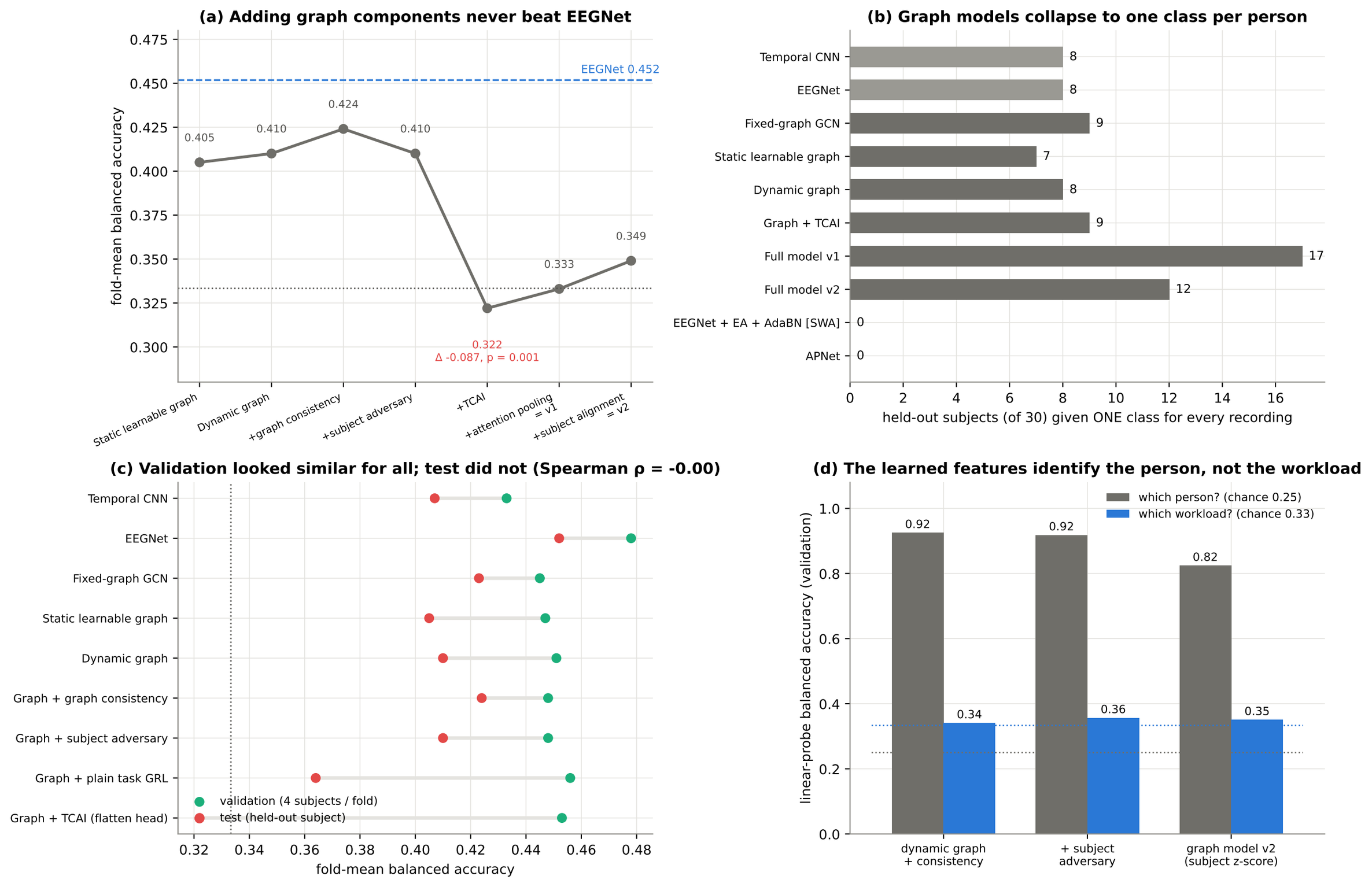
(a) Adding graph components never beat EEGNet
fold-mean balanced accuracy
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0.300
EEGNet 0.452
0.405
0.410
0.424
0.410
0.322
Δ -0.087, p = 0.001
0.333
0.349
Static learnable graph
Dynamic graph
+graph consistency
+subject adversary
+TCAI
+attention pooling = v1
+subject alignment = v2
(b) Graph models collapse to one class per person
Temporal CNN
EEGNet
Fixed-graph GCN
Static learnable graph
Dynamic graph
Graph + TCAI
Full model v1
Full model v2
EEGNet + EA + AdaBN [SWA]
APNet
8
8
9
7
8
9
17
12
0
0
0
2
4
6
8
10
12
14
16
held-out subjects (of 30) given ONE class for every recording
(c) Validation looked similar for all; test did not (Spearman ρ = -0.00)
Temporal CNN
EEGNet
Fixed-graph GCN
Static learnable graph
Dynamic graph
Graph + graph consistency
Graph + subject adversary
Graph + plain task GRL
Graph + TCAI (flatten head)
validation (4 subjects / fold)
test (held-out subject)
0.32
0.34
0.36
0.38
0.40
0.42
0.44
0.46
0.48
fold-mean balanced accuracy
(d) The learned features identify the person, not the workload
linear-probe balanced accuracy (validation)
which person? (chance 0.25)
which workload? (chance 0.33)
1.0
0.8
0.6
0.4
0.2
0.0
0.92
0.34
0.92
0.36
0.82
0.35
dynamic graph + consistency
+ subject adversary
graph model v2 (subject z-score)